

九. 杨-Mills理论与QCD

1. 杨-Mills 规范场

自由夸克场: u, d, s, c, t, b 6味

f: 1 2 3 4 5 6

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix}$$

↓ 每个 ψ_i 为 Dirac 旋量 (Fermi子)

3色

色转动: Gell-Mann 矩阵 U $SU(3)$

$$U = e^{i\theta_a \lambda^a / 2}, \text{tr } \lambda^a = 0, \text{tr}(\lambda^a \lambda^b) = 2\delta^{ab}$$

李代数: 结构常数 f_{abc} : 取 $T^a = \frac{\lambda^a}{2} \rightarrow U = e^{i\theta_a T^a}$, T^a 为李代数

$$[T^a, T^b] = if_{abc} T^c$$

$$\text{Jacobi: } [T^a, T^b] T^c + [T^b, T^c] T^a + [T^c, T^a] T^b = 0 \rightarrow f^{ade} f^{bcd} + f^{bde} f^{cad} + f^{cde} f^{abd} = 0$$

$$\text{对 } SU(3): f_{123}=1, f_{141}=-f_{151}=f_{246}=f_{257}=f_{345}=-f_{367}=\frac{1}{2}, f_{458}=f_{678}=\frac{\sqrt{3}}{2}$$

自由夸克场的拉氏密度: (仿 Dirac: $\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$) (质量与色无关)

色空间规范不变性: $\psi \rightarrow U\psi = e^{i\theta_a T^a} \psi$

整体 $\rightarrow m\bar{\psi}\psi = m\bar{\psi}\psi, \partial_\mu U = 0, \mathcal{L}$ 与色空间取向无关

定域 $\rightarrow \theta_a(x), m\bar{\psi}\psi = m\bar{\psi}U^\dagger U\psi = m\bar{\psi}\psi$

$$\partial_\mu \psi = \partial_\mu U \psi = U(\partial_\mu + U^\dagger \partial_\mu U)\psi$$

协变微分: $D_\mu = \partial_\mu + iqA_\mu$ (色空间的耦合常数与规范场) $\rightarrow$ 规范变换 $A'_\mu = UA_\mu U^\dagger + \frac{i}{q}(\partial_\mu U)U^\dagger$

各空间与闵空间的耦合

$$D'_\mu \psi' = D'_\mu U\psi = (\partial_\mu + iqA'_\mu)U\psi$$

$$= UA_\mu U^\dagger - \frac{i}{q}U\partial_\mu U^\dagger$$

$$= U(\partial_\mu + \underbrace{U^\dagger \partial_\mu U}_{A_\mu} + iqU^\dagger A'_\mu U)\psi = UD_\mu \psi \rightarrow \text{动能项不变: } \bar{\psi}i\gamma^\mu D'_\mu \psi' = \bar{\psi}i\gamma^\mu D_\mu \psi$$

$$\text{动能项 } (\bar{\psi}i\gamma^\mu D_\mu \psi)' = \bar{\psi}i\gamma^\mu D'_\mu \psi' = \bar{\psi}U^\dagger U i\gamma^\mu D_\mu \psi = \bar{\psi}i\gamma^\mu D_\mu \psi \rightarrow \text{不变}$$

定域规范不变的夸克场: $\mathcal{L}_\psi + \psi A = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi = \mathcal{L}_\psi + \mathcal{L}_\psi A$

规范场 A_μ

$-q\bar{\psi}\gamma^\mu A_\mu\psi$ 耦合项

$$U = \begin{cases} e^{i\gamma_5} \rightarrow U(1) \\ e^{i\vec{\theta} \cdot \vec{\sigma}} \rightarrow SU(2) \\ e^{i\theta^a T_a} \rightarrow SU(3) \end{cases}$$

六相关联

$$A_\mu^\dagger \rightarrow A_\mu'^\dagger = U A_\mu^\dagger U^\dagger - \frac{i}{q} U \partial_\mu U \rightarrow A_\mu' - A_\mu = U(A_\mu' - A_\mu)U^\dagger$$

$$\text{无穷小变换: } A_\mu' = A_\mu + i\theta^a [T^a, A_\mu] - \frac{i}{q} \partial_\mu \theta^a T^a \rightarrow \text{取迹 (tr } T^a = 0) \rightarrow \text{tr } A_\mu' = \text{tr } A_\mu \xrightarrow{\text{取初态}} \text{tr } A_\mu = 0$$

$$A_\mu = A_\mu^a T_a$$

$$A_\mu'^a = A_\mu^a - f_{abc} \theta^b A_\mu^c - \frac{1}{q} \partial_\mu \theta^a = A_\mu^a - \frac{1}{q} D_\mu^{ab} \theta^b$$

规范变化项

$$\text{协变导数 } D_\mu^{ab} = \delta^{ab} \partial_\mu + q f^{abc} A_\mu^c$$

拉氏量 (要求: Abelian 时可还原)

$$\mathcal{L}_A = -\frac{1}{2} \text{tr}(F_{\mu\nu} F^{\mu\nu}) = -\frac{1}{2} F_{\mu\nu}^a F_{\mu\nu}^b \text{tr}(T_a T_b) = -\frac{1}{4} F_{\mu\nu}^a F_{\mu\nu}^a$$

?

Maxwell 场: $D_\mu = \partial_\mu + iqA_\mu$

$$[D_\mu, D_\nu] = iqF_{\mu\nu} = iq(\partial_\mu A_\nu - \partial_\nu A_\mu) \rightarrow \text{定义 } F_{\mu\nu}$$

$$A_\mu \rightarrow A_\mu^a T_a \text{ 时, } \text{tr } A_\mu = 0 \rightarrow \text{tr } F_{\mu\nu} = 0 \rightarrow F_{\mu\nu} = F_{\mu\nu}^a T_a$$

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - q f^{abc} A_\mu^b A_\nu^c$$

$$F_{\mu\nu}'^a = U F_{\mu\nu} U^\dagger \xrightarrow{\text{变换}} (F_{\mu\nu}^a - f^{abc} \theta^b F_{\mu\nu}^c) T_a$$

协变项

附: 规范场的微分形式 (n-形式, 微分算符 d, 积分算符 \int)

$$F_{\mu\nu} \text{ 为 2-形式: } \begin{cases} \partial^\mu F_{\mu\nu} = j_\nu \\ \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0 \end{cases} \rightarrow \begin{cases} F = \frac{1}{2} F_{\mu\nu} dx^\mu dx^\nu = dA \\ dF = 0 \end{cases}$$

closed 封闭: 1-形式 α , $d\alpha = 0$

exact 平谨: 1-形式 α , $\exists \beta$ 使 $\alpha = d\beta$ (必封闭)

Poincaré 引理: $\xrightarrow{\text{封闭} \rightarrow \text{定域平谨}}$

简化: $iqA_\mu \rightarrow$ 写作 A_μ , $D_\mu = \partial_\mu + A_\mu$

$$1\text{-形式: } A = dx^\mu A_\mu \rightarrow A^2 = A_\mu A_\nu dx^\mu dx^\nu = \frac{1}{2} [A_\mu, A_\nu] dx^\mu dx^\nu$$

$$\text{规范变化: } A' = U A U^\dagger + U dU^\dagger, U \text{ 为 } 0\text{-形式} \rightarrow dU^\dagger = \partial_\mu U^\dagger dx^\mu$$

$$dA' = U dA U^\dagger + dU A U^\dagger - U A dU^\dagger + dU dU^\dagger$$

$$\text{同时: } A'^2 = U A^2 U^\dagger + U A dU^\dagger + U dU^\dagger U A U^\dagger + \underbrace{U dU^\dagger U A U^\dagger - dU A U^\dagger - U A dU^\dagger}_{=0} + U dU^\dagger U dU^\dagger$$

$$\rightarrow dA' + A'^2 = U dA U^\dagger + U A^2 U^\dagger = U (dA + A^2) U^\dagger \rightarrow F = A^2 + dA \text{ 协变}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$$

2. 杨-Mills 场路径积分

1. Faddeev-Popov 等式

量场中心等式 $\rightarrow Z[J] = \int DA e^{i \int dx (L + J A_\mu)} = \int DA e^{i S[A_\mu]}$

问题: 规范不变性 $\rightarrow$ 无穷个等效 $S \rightarrow$ 发散

解1: Feynmann 规范 (见7)

解2: Faddeev-Popov-De Witt 方法

★ 取超曲面为规范 $G^a[A_\mu] = 0$

条件数, 即自由度

目的: 同类场作用效果为1

使 $1 = \Delta[A_\mu] \int DU \delta[G[A_\mu^u]]$, $\delta[G[A_\mu^u]] = \prod_{a,x} \delta(G^a[A_\mu^u(x)]) \leftarrow \delta$ 泛函, $A_\mu^u \rightarrow A'_\mu = u A_\mu$

2次变换: $u' = u' u \rightarrow 1 = \Delta[A_\mu^u] \int DU' \delta[G[A_\mu^{u'}]] \rightarrow \Delta[A_\mu^u] = \Delta[A_\mu]$, 规范不变
规范不变 $\rightarrow \delta G A_\mu^u$

F-P 行列式 $\Delta[A_\mu]$ (类似 Jacobi, 归一化用)

$$\text{对 } u = e^{i\theta T^a} \rightarrow 1 = \Delta[A_\mu] \int D\theta \delta(G[A_\mu^\theta]) = \int \prod_{a,y} d\theta^a \delta(G^a[A_\mu^\theta]) = \int \frac{\delta(\theta^1_y, \theta^2_y, \dots)}{\delta(\theta^1_y, \theta^2_y, \dots)} \prod_{a,y} d\theta^a \delta G^a[A_\mu^\theta]$$

$$= \frac{\delta(\theta^1, \dots)}{\delta(\theta^1_y, \dots)} \Big|_{G^a[A_\mu] = 0}$$

$$\Delta A_\mu = \frac{\delta(G^1(x), \dots)}{\delta(\theta^1_y, \dots)} \Big|_{G^a[A_\mu] = 0}$$

拉氏密度 $Z[J] = \int DA DU \Delta[A_\mu] \delta[G^a[A_\mu^u]] e^{i S[A_\mu]}$

↓ 规范不变, $A_\mu \rightarrow A_\mu^u$, 使 DU 分离 $\rightarrow$ 进入归一化系数

$$Z[J] = \int DA \Delta[A_\mu] \delta[G[A_\mu]] e^{i S[A_\mu]}$$

规范固定项: 把 δ 积分: 取 $G^a[A_\mu] = \Omega^a[A_\mu] - w_\mu^a = 0 \rightarrow \delta G[A_\mu] = \Omega^a \cdot \delta A_\mu(x) - \delta w_\mu^a(x)$

$$Z[J] = Z[J] \int D\omega e^{-i \int dx \omega^a w_\mu^a / 25} = \int DA \Delta[A_\mu] e^{i S[A_\mu] - i \int dx \omega^a w_\mu^a / 25}$$

见5.1, $= 5$, 被归一化, 不考虑

规范固定项, $\Omega^2 = \int dx \Omega^a(x) \Omega_a(x)$

eg. Lorentz 规范: $\Omega^a = \partial^\mu A_\mu^a$

columb : $\nabla \cdot A^a$

轴向规范: A_3^a

瞬时规范: A_0^a

七章: 规范固定项: $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{GF}$
 (来源: 自然导出) $\frac{1}{2}(\partial_\mu A^\mu)^2$

$\rightarrow \frac{1}{2} = \lambda, \frac{\Omega^2}{2s}$ 为规范固定的逻辑推演结果

鬼场: $\Delta[A_\mu] = \underbrace{[\det \frac{\partial G_{xi}}{\partial \theta_i}]}_{iM}$

用于消除多余自由度的辅助粒子

继续, 使鬼场规范

Grassmann 泛函: $\int D\bar{\eta} D\eta e^{-\bar{\eta} K \eta} = \det K$

$\mathcal{L}_{FPG} = -\bar{\eta} M \eta$ (鬼场拉氏量)

类时态与纵向极化

$M_{(x,y)}^{ab} = -q \frac{\delta G^a(x)}{\delta \theta^b(y)} = -q \frac{\delta \Omega^a(x)}{\delta \theta^b(y)} = -q \int dz \frac{\delta \Omega^a(x)}{\delta A_\mu^c(z)} \frac{\delta A_\mu^c(z)}{\delta \theta^b(y)} = \int dz \frac{\delta \Omega^a(x)}{\delta A_\mu^c(z)} D_\mu^{cb}(z) \delta(z-y)$

杨-Mills 场传播子

Lorentz: $\delta^{ac} \partial_x^\mu \delta(x-z)$

规范项: $\mathcal{L}_{GF} = -\frac{1}{2s} (\partial^\mu A_\mu^a) (\partial^\nu A_\nu^a) = \frac{1}{2s} A_\mu^a \partial_\mu \partial^\nu A_\nu^a$

鬼场: $\mathcal{L}_{FPG} = -\bar{\eta} M \eta$, $M_{(x,y)}^{ab} = \int dz [\delta^{ac} \partial_x^\mu \delta(x-z) D_\mu^{cb}(z) \delta(z-y)] = \partial_x^\mu D_\mu^{ab}(x) \cdot \delta(x-y)$

$= -\int dx dy \bar{\eta}(x) M_{(x,y)}^{ab} \eta(y) = -\int dx \bar{\eta} \partial^\mu [D_\mu^{ab}(x) \eta(x)] = \int dx (\partial^\mu \bar{\eta}) (D_\mu^{ab}(x) \eta) = 0$
 $D_\mu^{ab}(x) = (\delta^{ab} \partial_\mu + q f^{abc} A_\mu^c)$
 $= -\int dx \bar{\eta}_a \square \eta^a - q \int dx f^{abc} (\partial^\mu \bar{\eta}_a A_\mu^b \eta_c)$

传播子: $i\tilde{D}_F^{ab}(x-y) \rightarrow \tilde{D}_F^{ab}(x-y) = \frac{1}{(2\pi)^4} \int dk \tilde{D}_F^{ab}(k) e^{-ik(x-y)}$
 $\tilde{D}_F^{ab}(k) = \delta^{ab} \frac{1}{k^2}$
 $= \underbrace{-\bar{\eta} \square \eta}_{\text{传播子}} - \underbrace{q(\partial^\mu \bar{\eta}) [A_\mu, \eta]}_{\text{与规范场耦合}}$

规范场: $\mathcal{L}_A = -\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a} = -\frac{1}{4} (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a - q f^{abc} A_\mu^b A_\nu^c) \cdot C.C$

$= -\frac{1}{4} (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a) \cdot C.C + \frac{1}{2} q f^{abc} A_\mu^b A_\nu^c (\partial^\mu A_\alpha^a - \partial^\nu A_\alpha^a) - \frac{1}{4} q^2 f^{abc} f^{abc'} A_\mu^b A_\nu^c A_\alpha^a A_\alpha^{c'}$
 $\rightarrow \frac{1}{2} A_\mu^a (g^{\mu\nu} \square - \partial^\mu \partial^\nu) A_\nu^a$
 $q f^{abc} A_\mu^b A_\nu^c \partial^\mu A_\alpha^a$

加入规范项, $\frac{1}{2} A_\mu^a (g^{\mu\nu} \square - (1-\lambda) \partial^\mu \partial^\nu) A_\nu^a$

传播子: $iD_F^{ab}(x-y)$, $D_F^{ab}(k) = -\delta^{ab} \frac{1}{k^2} (g^{\mu\nu} - (1-s) \frac{k^\mu k^\nu}{k^2})$

轴向规范: $t^\mu A_\mu^a = 0$, $t^\mu t_\mu = -1$ (类空)

$$\Omega^a = t^\mu A_\mu^a, \quad L_{GF} = -\frac{1}{2\xi} (t^\mu A_\mu^a)^2 = \frac{1}{2\xi} A_\mu^a t^\mu t^\nu A_\nu^a,$$

$$M_{(c,y)}^{ab} = -t^\mu D_\mu^{ab}(x) S(x-y) \rightarrow L_{FPQ} = \int dx \bar{\eta}_a(x) \underbrace{t^\mu D_\mu^{ab}(x)}_{\delta_\mu^\nu + g f^{abc} A_\mu^c} \eta_b(x) = \bar{\eta} \frac{\partial}{\partial \bar{\eta}} \eta + 0$$

轴向规范鬼场为自由场 $\rightarrow$ 积分到归一化常数 $\rightarrow$ 可考虑

传播子 $iD_{F\mu\nu}^{ab}(x-y) = i\delta^{ab} (g^{\mu\nu} \square - \gamma^\mu \gamma^\nu - \frac{1}{\xi} t^\mu t^\nu)^{-1}$ (更复杂)

3. QCD Feynmann 规则 Quantum Chromodynamics

$i\sigma T_a \rightarrow SU(2)$, 色空间, 规范场

$i\sigma \cdot \theta \rightarrow SU(2)$, 旋电同

$i\gamma(x) \rightarrow U(1)$, A_μ , 电磁场

quark confinement 夸克禁闭: 未发现单夸克 $\rightarrow$ 夸克以单色或无色出现
color confinement

渐近自由: 距离 $\uparrow$, 相互作用 $\downarrow$

$$c g = g$$

QCD 拉氏密度: $\mathcal{L} = \mathcal{L}_\psi + \mathcal{L}_{\psi A} + (\mathcal{L}_A + \mathcal{L}_{GF}) + \mathcal{L}_{FPQ}$

规范场-夸克
规范场
规范场自耦合
鬼场

$$-g f^{abc} (\bar{\psi} \gamma_\mu A_\mu^b) \psi_c$$

$$- \bar{\eta}_a \square \eta^a$$

$$\frac{1}{2} A_\mu^a [g^{\mu\nu} \square - (1-\lambda) \gamma^\mu \gamma^\nu] A_\nu^a$$

胶子顶点:

$$-g \bar{\psi} \gamma^\mu A_\mu \psi$$

$$\bar{\psi}(i\not{\partial} - m)\psi$$

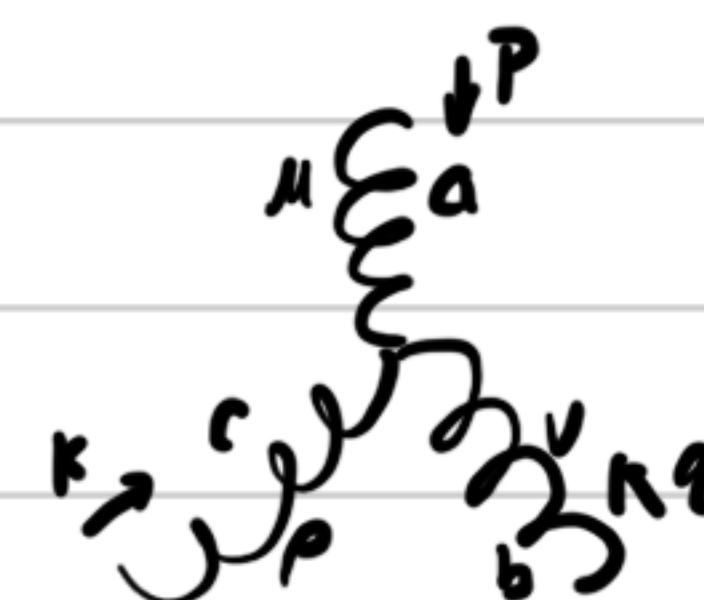
$$-ig\gamma^\mu T_a$$

$$iS_F^m(p)$$

$$\text{三胶子顶点: } -gf^{abc} A_\mu^a A_\nu^b \partial^\mu A_\nu^c$$

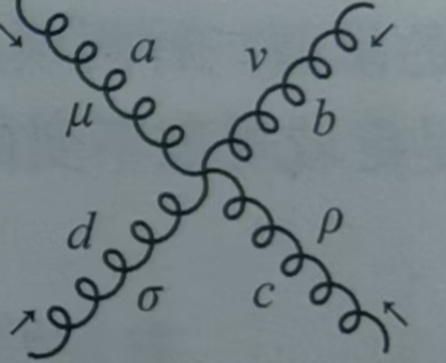
$$-gf^{abc} [g^{\mu\nu}(p-q)^\rho + g^{\nu\rho}(q-k)^\mu + g^{\rho\mu}(k-p)^\nu]$$

$$\text{四胶子顶点: } -ig^2 f^{abc} f^{a'b'c'} A_\mu^b A_\nu^c A_\mu^{a'} A_\nu^{b'}$$



4. BRST 对称

规范不变性 $\leftrightarrow$ BRST 对称性

$$\left. \begin{aligned} & -ig^2[f^{abe}f^{cde}(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\sigma}g^{\nu\rho}) \\ & + f^{ace}f^{bde}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\sigma}g^{\rho\nu}) \\ & + f^{ade}f^{bce}(g^{\mu\nu}g^{\sigma\rho} - g^{\mu\rho}g^{\sigma\nu})] \end{aligned} \right\} =$$


BRST 变换: 对协变规范: $\bar{\psi}(i\not{D}-m)\psi - \frac{1}{4}(F_{\mu\nu}^a)^2 - \frac{1}{2\xi}(\partial^\mu A_\mu^a)^2 - \bar{\eta}_b \partial^\mu D_\mu^{ab} \eta^b$

引入 $\theta^a = -g\epsilon\eta^a$ (ϵ 为 Grassmann 数, 使 θ 为 Grassmann 数 (η 为 Grassmann 数), $\epsilon\eta + \eta\epsilon = 0$, $\epsilon\psi + \psi\epsilon = 0$)

$$U = e^{i\theta^a T_a} \rightarrow \delta A_\mu^{ab} = \epsilon D_\mu^{ab} \eta^b$$

$$\delta\psi = -ig\epsilon\eta_a T^a \psi$$

$$\rightarrow \mathcal{L} \text{ 不变, } \delta\eta^a = \frac{1}{2}g\epsilon f^{abc}\eta^b\eta^c$$

$$\delta\bar{\eta}^a = -\frac{\epsilon}{\xi}\partial^\mu A_\mu^a$$

$$\delta\mathcal{L} = -\delta\left(\frac{1}{2\xi}(\partial^\mu A_\mu^a)^2 + \bar{\eta}_b \partial^\mu D_\mu^{ab} \eta^b\right)$$

$$= -\frac{1}{\xi}\partial^\mu A_\mu \delta\partial_\mu + \delta\bar{\eta}_b \partial^\mu D_\mu^{ab} \eta^b + \bar{\eta}_b \partial^\mu \delta(D_\mu^{ab} \eta^b)$$

$$\delta(D_\mu^{ab} \eta^b) = D_\mu^{ab} \delta\eta^b + g f^{abc} \eta^b \delta A_\mu^c = D_\mu^{ab} \delta\eta^b + g f^{abc} \eta^b \delta A_\mu^c = g\epsilon\left(\frac{1}{2}f^{bcd} D_\mu^{ab} \eta^d - f^{bcd} \eta^b D_\mu^{cd} \eta^d\right)$$

计算太长, 不写了!

$$= \frac{1}{2}g(f^{ade}f^{bcd} + f^{bde}f^{acd} + f^{cde}f^{abd}) A_\mu^e \eta^b \eta^c$$

Jacobi

算符: $\delta\phi = \epsilon Q\phi$,

算子

$Q^2 = 0$

$$QA_\mu^a = D_\mu^{ab} \phi^b, Q^2(A_\mu^a) = \frac{1}{2}\delta(D_\mu^{ab} \phi^b) = 0$$

$$Q\psi = -ig\eta^a T^a \psi, Q^2\psi = -ig^2\left(\frac{1}{2}f^{abc}\eta^b\eta^c - i\eta_b T^b \eta^a\right) T^a \psi = 0$$

$$Q^2\bar{\eta}^a = -\frac{1}{\xi}\delta(D_\mu^{ab} \eta^b), \text{全微分, 视 } 0$$

$$Q^2\eta^a = \frac{g^3}{8}f^{abc}f^{bde}\eta^d\eta^e f^{cfa}\eta^f\eta^g =$$

生成泛函体现: 加入源 u, v : $\mathcal{L} = \mathcal{L}_{FP} + s^\mu A_\mu + \chi^a \eta^a + \bar{\eta}^a y^a + u^a (D_\mu^{ab} \eta^b) + v^a (\frac{1}{2}g f^{abc} \eta^b \eta^c)$

$$Z[s, \chi, y, u, v] = \int DA_\mu D\bar{\eta} D\eta e^{iS}$$

$$\delta\mathcal{L} = s^\mu \delta A_\mu + \chi^a \delta\eta^a + \delta\bar{\eta}^a y^a \rightarrow \text{Jacobi: } J = \left| \frac{\delta[A_\mu^b + \delta A_\mu^b, \bar{\eta}^b + \delta\bar{\eta}^b, \eta^b + \delta\eta^b]}{\delta[A_\mu^b(y), \bar{\eta}^b(y), \eta^b(y)]} \right|$$

非0元: $\frac{\delta[A+\delta A]}{\delta A} = \delta^{ab} \delta_{\mu\nu} \delta(x-y) + \frac{\epsilon \delta[D_\mu^a \xi_\nu] \eta^c(x)}{\delta A_\nu^b(y)} = \delta_{\mu\nu}^{ab} \delta(x-y) + g \epsilon f^{acd} \eta^c(x) \delta_{\mu\nu}^{bd} \delta(x-y) = \delta_{\mu\nu}^{ab} \delta(x-y) (\delta^{ab} - g \epsilon f^{abc} \eta^c)$

$\frac{\delta[A+\delta A]}{\delta \eta^b} = \frac{\epsilon \delta[D_\mu^a \xi_\nu] \eta^c(x)}{\delta \eta^b(y)} = \epsilon D_\mu^{ab} \delta(x-y) = \epsilon \partial_\mu^x \delta_{\mu\nu}^{ab} \delta(x-y) + g f^{abc} A_\mu^c \delta(x-y)$

$\frac{\delta[\bar{\eta}^a + \delta \bar{\eta}]}{\delta A_\nu^b} = -\delta^{ab} \partial_x^\nu \delta(x-y) \frac{\epsilon}{\xi}$

$\frac{\delta[\bar{\eta}^a + \delta \bar{\eta}]}{\delta \bar{\eta}} = \delta^{ab} \delta(x-y)$

$\frac{\delta[\eta^a + \delta \eta]}{\delta \eta} = \delta^{ab} \delta(x-y) + \delta(x-y) g \epsilon f^{abc} \eta^c = \delta(x-y) (\delta^{ab} + g \epsilon f^{abc} \eta^c)$

相加不含 ϵ -次项

ϵ 为 q 数, $\epsilon^2=0 \rightarrow 2$ 个为 0

故有: $\det J = \prod J_{ii} = (\delta_{\mu\nu})^M [\delta(x-y)]^N$

★不变 $\rightarrow$ BRST 变换保测

$\downarrow$
 $Z \rightarrow Z' = Z$

非对角元正比于 ϵ , $\epsilon=0 \rightarrow$ 不参与

生成泛函的 B 不变的条件: $\delta Z[s, x, y, u, v] = i \int D A D \bar{\eta} D \eta e^{iS} \int dx (s^{\mu a} \delta A_\mu^a + x^a \delta \eta^a + \delta \bar{\eta}^a y^a)$
 $= i \epsilon \int dx [s^{\mu a} \frac{\delta Z}{\delta u^{\mu a}} + x^a \frac{\delta Z}{\delta v^a} - \frac{1}{\xi} (\partial^\mu \frac{\delta Z}{\delta s^{\mu a}}) y^a] = 0$

对于 $Z = e^{iW}$, (取 Green)

Green 函数的 Slavnov-Taylor 恒等式: $\int dx [s^{\mu a} \frac{\delta W}{\delta u^{\mu a}} + x^a \frac{\delta W}{\delta v^a} - \frac{1}{\xi} (\partial^\mu \frac{\delta W}{\delta s^{\mu a}}) y^a] = 0$

Legendre

顶角方程: $\Gamma[A_\mu, \bar{\eta}, \eta; u, v] = W[s, x, y, u, v] - \int dx (s^{\mu a} A_\mu^a + x^a \eta^a + \bar{\eta}^a y^a)$

代入: $\int dx [\frac{\delta \Gamma}{\delta A_\mu^a} \frac{\delta \Gamma}{\delta u^{\mu a}} + \frac{\delta \Gamma}{\delta \eta^a} \frac{\delta \Gamma}{\delta v^a} - \frac{1}{\xi} (\partial^\mu A_\mu^a) \frac{\delta \Gamma}{\delta \bar{\eta}^a}] = 0$

$\int dx [(\frac{\delta \Gamma}{\delta A_\mu^a} - \frac{1}{\xi} (\partial^\mu \partial^\nu A_\nu^a)) \frac{\delta \Gamma}{\delta u^{\mu a}} + \frac{\delta \Gamma}{\delta \eta^a} \frac{\delta \Gamma}{\delta v^a}] = 0 \rightarrow$ 取 $\Gamma' = \Gamma + \frac{1}{2\xi} \int dx (\partial^\mu A_\mu^a)^2$

$\int dx [\frac{\delta \Gamma'}{\delta u^{\mu a}} \frac{\delta \Gamma'}{\delta A_\mu^a} + \frac{\delta \Gamma'}{\delta v^a} \frac{\delta \Gamma'}{\delta \eta^a}] = 0$ (微分几何联系) ?

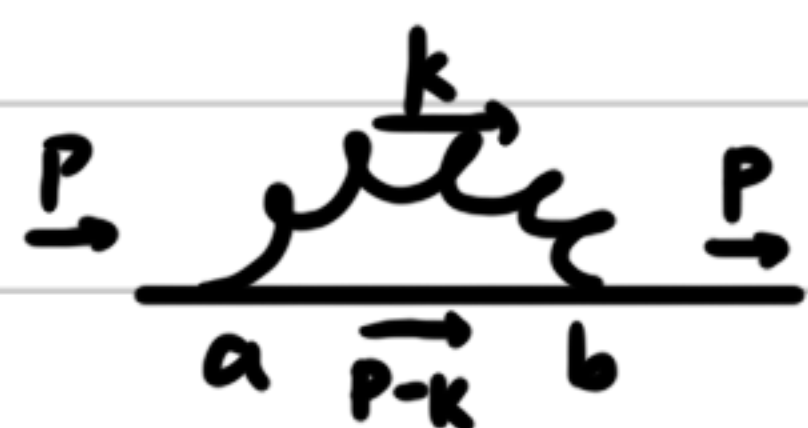
同时:

$\frac{\delta Z}{\delta \bar{\eta}^a} = \int D A_\mu D \bar{\eta} D \eta i (y^a - \partial^\mu D_\mu^{ab} \eta^b) e^{iS} = i y^a Z - \partial^\mu \frac{\delta Z}{\delta u^{\mu a}}$

$\Rightarrow \frac{\delta W}{\delta \bar{\eta}^a} = y^a - \partial^\mu \frac{\delta W}{\delta u^{\mu a}} = 0$

$\frac{\delta \Gamma}{\delta \bar{\eta}^a} = -\partial^\mu \frac{\delta \Gamma}{\delta u^{\mu a}}$

5. 单圈重正化



夸克自能 $-i\Sigma(p) = -i\mu^{4-d}g^2 \int \frac{d^d k}{(2\pi)^d} (\gamma^\mu T_a) iD_{F\mu\nu}(k) iS_F(p-k) (\gamma^\nu T_b)$

分离: $\Sigma(p) = T^a T_a \cdot \Sigma_{\text{qed}}(p)$ 后者 $= i\mu^{4-d}g^2 \int \frac{d^d k}{(2\pi)^d} \gamma^\mu D_{F\mu\nu}(k) S_F(p-k) \gamma^\nu = \frac{\alpha_s}{2\pi\epsilon}(-\not{p} + 4m)$, $\alpha_s = \frac{g^2}{4\pi}$

$(T^a T_a)_{mn} = C_1 \delta_{mn} = \frac{1}{4}(\lambda_1^2 + \dots + \lambda_8^2) = \frac{4}{3} \Rightarrow \Sigma(p) = \frac{2\alpha_s}{3\pi\epsilon}(-\not{p} + 4m)$

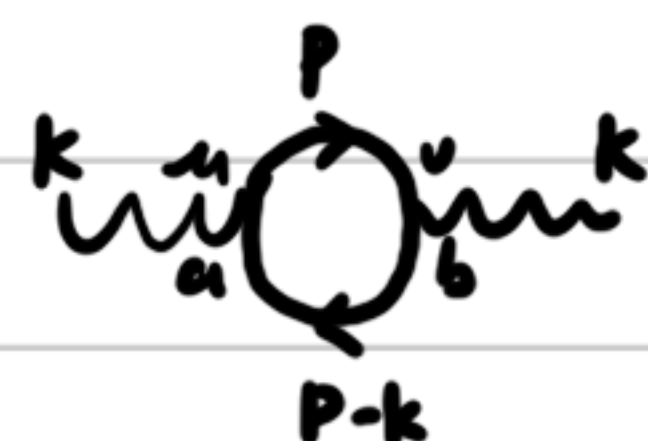
$Z_2 = 1 - \frac{2\alpha_s}{3\pi\epsilon}$, $Z_m = 1 - \frac{8\alpha_s}{3\pi\epsilon}$

→ QCD重正化对同色夸克有贡献, $D_{F\mu\nu}^{ab}(k)$ 在 $a=b$ 时非0 → 胶子线两端同色

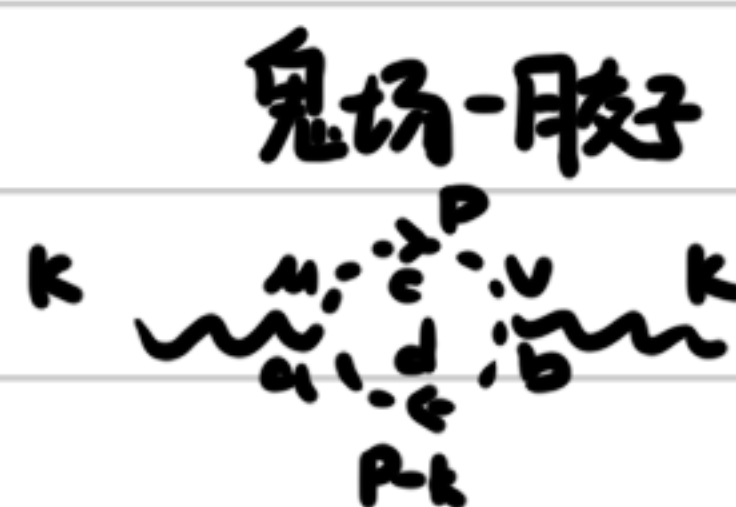
真空极化: 胶子完全传播子

$$\begin{aligned} & \text{真空极化} \\ & = \text{胶子-胶子} + \text{胶子-鬼场} \\ & + \text{胶子-夸克-反夸克} + \text{胶子-四夸克} + \dots \end{aligned}$$

胶子-胶子: $i\Pi_{\mu\nu}^{ab}(k) = \mu^{4-d}g^2 \int \frac{d^d p}{(2\pi)^d} \text{tr}[\gamma_\mu T^a iS_F(p) \gamma_\nu T^b iS_F(p-k)]$

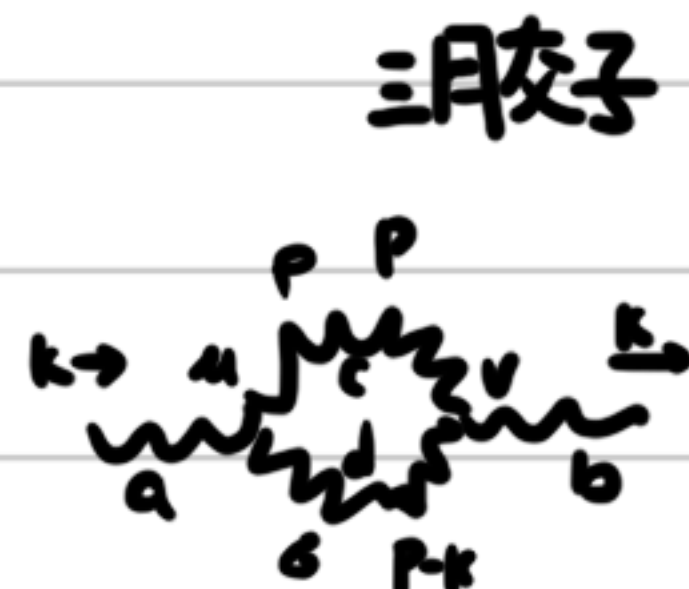


$= \text{tr}[T^a T^b] \cdot i\Pi_{\mu\nu}^{\text{qed}}(k)$ 后者 $\Pi_{\mu\nu}^{\text{qed}}(k) = -i\mu^{4-d}g^2 \int \frac{d^d p}{(2\pi)^d} \text{tr}[\gamma_\mu iS_F(p) \gamma_\nu iS_F(p-k)] = -\frac{2\alpha_s}{3\pi\epsilon}(g_{\mu\nu}k^2 - k_\mu k_\nu)$
 $= -\frac{nf}{2}\delta^{ab} i\frac{2\alpha_s}{3\pi\epsilon}(g_{\mu\nu}k^2 - k_\mu k_\nu)$ nf 为可激发的夸克味数



鬼场-胶子

$$\begin{aligned} i\Pi_{\mu\nu}^{ab}(k) &= \mu^{4-d}g^2 \int \frac{d^d p}{(2\pi)^d} f^{cad} p_\mu i\bar{D}_F(p) f^{dbc} (p-k)_\nu i\bar{D}_F(p-k) \\ &= -\mu^{4-d}g^2 f^{cad} f^{dbc} \int \frac{d^d p}{(2\pi)^d} \frac{p_\mu (p-k)_\nu}{p^2 (p-k)^2} \xrightarrow{\text{积分}} \int \frac{d^d p}{(2\pi)^d} \frac{p_\mu (p-k)_\nu}{p^2 (p-k)^2} = -\frac{i}{16\pi\epsilon}(\frac{1}{6}g_{\mu\nu}k^2 + \frac{1}{3}k_\mu k_\nu) \\ &= i\frac{C_2\alpha_s}{4\pi\epsilon}(\frac{1}{6}g_{\mu\nu}k^2 + \frac{1}{3}k_\mu k_\nu) \end{aligned}$$



三胶子

$$i\Pi_{\mu\nu}^{ab}(k) = -\frac{1}{2}\mu^{4-d}g^2 f^{acd} f^{bcd} \int \frac{d^d p}{(2\pi)^d} \frac{F_{\mu\nu}}{p^2 (p-k)^2}$$
 $F_{\mu\nu} = [g_{\mu\rho}(k+p)_\nu - g_{\rho\nu}(-p-p+k)_\mu + g_{\rho\mu}(p-k-k)_\rho] \times [g_\rho{}^\sigma(-k-p)_\sigma + g^\rho{}^\sigma(p-p-k)_\nu + g^\sigma{}_\nu(-p+k-k)_\rho]$

$\int g_{\mu\rho}g^{\rho\sigma} = g^2\delta = d$

$\Pi_{\mu\nu}^{ab}(k) = \frac{C_2\alpha_s}{4\pi\epsilon}(\frac{19}{6}g_{\mu\nu}k^2 - \frac{11}{3}k_\mu k_\nu)\delta^{ab}$

在 $d=2$ 时积分 $\int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2}$ 发散, 归到00项

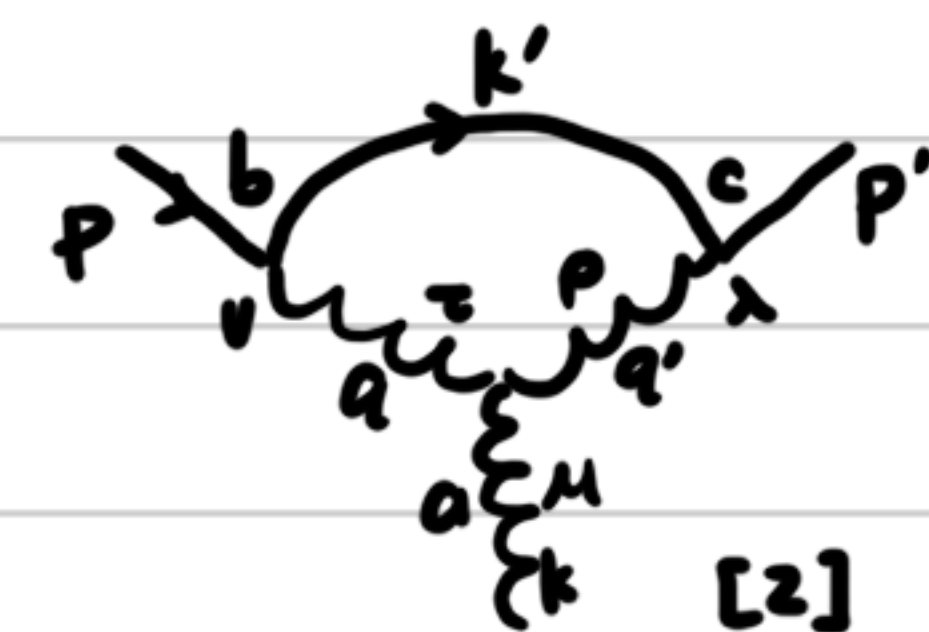
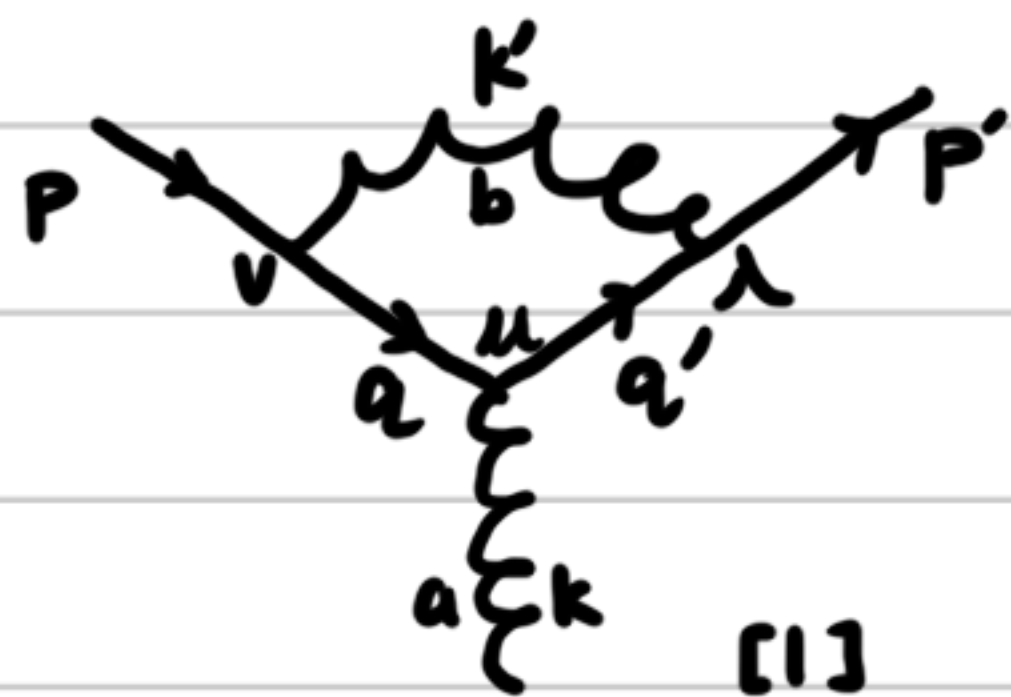


四胶子

线上: $\Pi_{\mu\nu}^{ab}(k) = (5 - \frac{2nf}{3})\frac{\alpha_s}{2\pi\epsilon}(g_{\mu\nu}k^2 - k_\mu k_\nu)\delta^{ab}$

$Z_3 = 1 + (5 - \frac{2nf}{3})\frac{\alpha_s}{2\pi\epsilon}$

夸克-胶子顶角:



$$\text{顶角函数 } \Lambda_{[1]}^{a\mu}(P', P) = i\mu^{4-d} g^2 \int \frac{d^d k'}{(2\pi)^d} \gamma^\nu T^b S_F(q) \gamma^\mu T^a S_F(q') \gamma^\lambda T^b D_{F\lambda\nu}(k')$$

$$= T^b T^a T^b \cdot \Lambda_{QED}^\mu(P', P)$$

$$T^b T^a T^b = T^b (T^b T^a + i f^{abc} T^c) = (C_1 - \frac{1}{2} C_2) T^a$$

$$\text{故 } \Lambda_{[1]}^{a\mu}(P', P) = (C_1 - \frac{1}{2} C_2) \frac{g_s}{2\pi\epsilon} \gamma^\mu T^a$$

$$\Lambda_{[2]}^{a\mu}(P', P) = \mu^{4-d} g^2 \int \frac{d^d k'}{(2\pi)^d} \gamma^\lambda T^c S_F(k') \gamma^\nu T^b D_{\lambda\rho}(q') \cdot f^{abc} [g^{\mu\nu}(k-q)^\rho + g^{\nu\rho}(q+q')^\mu + g^{\rho\mu}(-q'-k)^\nu] D_{\tau\nu}(q)$$

$$= -\mu^{4-d} g^2 f^{abc} T^b T^c I^\mu$$

$$I^\mu = \int \frac{d^d k'}{(2\pi)^d} \frac{\gamma_\lambda (k' + m) \gamma_\nu k^{\lambda\mu\nu}}{(k'^2 - m^2)(P' - k')^2 (P - k')^2}$$

↓
仅考虑 k' 2次及以上项

$$\gamma_\lambda k'_\nu (g^{\mu\nu} k'^\lambda - g^{\nu\lambda} 2k'^\mu + g^{\lambda\mu} k'^\nu) = 2k'^2 \gamma^\mu + 2(d-2) k'_\nu k'^\mu$$

$$I^\mu = \int \frac{d^d k'}{(2\pi)^d} \frac{2k'^2 \gamma^\mu + 2(d-2) k'_\nu k'^\mu}{(k'^2 - m^2)(P' - k')^2 (P - k')^2}$$

$$\text{拆项公式: } \frac{1}{abc} = 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{(a(1-x-y) + bx + cy)^3}$$

$$I^\mu = 2 \int dx dy \int \frac{d^d k}{(2\pi)^d} \frac{2k^2 \gamma^\mu + 2(d-2) k k^\mu}{[k^2 - m^2(1-x-y) + P'^2 x^2 + P^2 y^2 - (P'x + Py)^2]^3}$$

$$= 2 \int_0^1 dx \int_0^{1-x} dy \frac{i}{(4\pi)^{\frac{d}{2}}} \frac{\gamma^\mu}{[m^2(1-x-y) - P'^2 x^2 + P^2 y^2 + (P'x + Py)^2]^{2-d/2}}$$

$$= \frac{i}{(4\pi)^2} \frac{6}{\epsilon} \gamma^\mu$$

$$\text{故 } \Lambda_{[2]}^{a\mu} = -\frac{1}{2} C_2 \mu^{4-d} g^2 T^a I^\mu = C_2 \frac{3g_s}{4\pi\epsilon} \gamma^\mu T^a$$

$$\text{总顶角 } \Lambda^{a\mu} = (C_1 + C_2) \frac{g_s}{2\pi\epsilon} \gamma^\mu T^a = \frac{13}{3} \frac{g_s}{2\pi\epsilon} \gamma^\mu T^a$$

$$Z_1 = 1 - \frac{13}{3} \frac{g_s^2}{2\pi\epsilon}$$

$$k^{\mu\nu\lambda} = [g^{\mu\nu}(k-q)^\lambda + g^{\nu\lambda}(q+q')^\mu + g^{\lambda\mu}(-q'-k)^\nu]$$

$$\left. \begin{array}{l} q = P - k' \\ q' = P' - k' \\ k = q' - q \end{array} \right\} \rightarrow = [g^{\mu\nu}(-2P + P' + k')^\lambda + g^{\nu\lambda}(P + P' - 2k')^\mu + g^{\lambda\mu}(P - 2P' + k')^\nu]$$

6. 重整化群

维数正规化中 μ 含质量量纲

标度变换: $\mu \rightarrow e^S \mu$, 未重整化的顶角 Γ_B 在该变换不变

重整化群, μ 为重整化标度因子

$$\text{正规顶角 } \Gamma(x_1 \dots x_n) = \frac{\delta^n \Gamma(\phi)}{\delta \phi(x_1) \dots \delta \phi(x_n)} \Big|_{\phi=\phi_c} \Rightarrow \text{动量 } \Gamma(p_1 \dots p_n; g, m, \mu) = Z_2^{F/2} Z_3^{B/2} \Gamma_B(p_1 \dots p_n, g_B, m_B)$$

$F+B=n$, 为 Fermi/Bose 子外线数

$$\text{两边 } \mu \frac{d}{d\mu}: \left(\mu \frac{\partial}{\partial \mu} + \mu \frac{\partial g}{\partial \mu} \frac{\partial}{\partial g} + \mu \frac{\partial m}{\partial \mu} \frac{\partial}{\partial m} \right) \Gamma = \left(\frac{F\mu}{2Z_2} \frac{\partial Z_2}{\partial \mu} + \frac{B\mu}{2Z_3} \frac{\partial Z_3}{\partial \mu} \right) \Gamma$$

$$\text{定义: } \beta = \mu \frac{\partial g}{\partial \mu}, \quad \gamma_m = \frac{\mu}{m} \frac{\partial m}{\partial \mu}, \quad \gamma_F = \frac{\mu}{Z_2} \frac{\partial Z_2}{\partial \mu}, \quad \gamma_B = \frac{\mu}{Z_3} \frac{\partial Z_3}{\partial \mu}$$

(都为 g 的函数)

$$\text{重整化群方程: } \left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} + m \gamma_m \frac{\partial}{\partial m} - F \gamma_F - B \gamma_B \right) \Gamma = 0$$

(又作 Callan-Symanzik 方程)

$$\text{标度行为: } \delta(x-y) = \frac{\delta \phi(x)}{\delta \phi(y)}$$

$$\left[\frac{\delta}{\delta \psi} \right] = \left[\delta(x-y) \right] [\psi]^{-1} = \Lambda^d \Lambda^{\frac{d-1}{2}} = \Lambda^{\frac{d+1}{2}} \rightarrow \Gamma(x_1 \dots x_n) \text{ 动量幂次 } D' = \frac{1}{2} F(d+1) + \frac{1}{2} B(d+2)$$

$$\left[\frac{\delta}{\delta \Lambda} \right] = \Lambda^d \Lambda^{\frac{d-1}{2}} = \Lambda^{\frac{d+2}{2}}$$

$\Gamma(p_1 \dots p_n)$ 为 $\Gamma(x)$ 的傅里叶变换:

$D = D' - nd + d$, 为正规顶角的正则量纲

$$\text{当 } p_i \rightarrow \xi p_i, \quad \Gamma(\xi p_i, g, m, \mu) = \Gamma(\xi p_i, g, \xi \xi' m, \xi \xi' \mu) \quad m \text{ 与 } \mu \text{ 有动量量纲}$$

$$= \xi^D (p_i, g, \xi' m, \xi' \mu)$$

$$\left(\xi \frac{\partial}{\partial \xi} + m \frac{\partial}{\partial m} + \mu \frac{\partial}{\partial \mu} - D \right) \Gamma(\xi p_i, g, m, \mu) = 0$$

$$\text{代入重整化群方程: } \left[-\xi \frac{\partial}{\partial \xi} + \beta \frac{\partial}{\partial g} + m(\gamma_m - 1) \frac{\partial}{\partial m} - F \gamma_F - B \gamma_B + D \right] \Gamma(\xi p_i, g, m, \mu) = 0$$

无相互作用时: $\beta, \gamma_m, \gamma_F, \gamma_B$ 为 0 $\Rightarrow$ 仅正则量纲作用

有相互作用: 微扰反常量纲, 根源: 量子理论重整化

Γ 的形式解: 猜解 $\Gamma = f(\xi) \Gamma(p, g(\xi), m(\xi), \mu)$, 求 $\frac{\partial}{\partial \ln \xi}$ 得:

$$\xi \frac{\partial}{\partial \xi} \Gamma(\xi p, g, m, \mu) = \left(\xi \frac{\partial f}{\partial \xi} + f \xi \frac{\partial g}{\partial \xi} \frac{\partial}{\partial g} + f \xi \frac{\partial m}{\partial \xi} \frac{\partial}{\partial m} \right) \Gamma(p, g(\xi), m(\xi), \mu)$$

$$\downarrow$$

$$\frac{1}{f} \Gamma(\xi p, g, m, \mu)$$

$$\left(-\xi \frac{\partial}{\partial \xi} + \xi \frac{df}{d\xi} + \xi \frac{\partial g}{\partial \xi} \frac{\partial}{\partial g} + \xi \frac{\partial m}{\partial \xi} \frac{\partial}{\partial m} \right) \Gamma(\xi p, g, m, \mu) = 0$$

与上式对比: $\beta(g) = \xi \frac{\partial g}{\partial \xi}$ $m[\gamma_m(g)-1] = \xi \frac{\partial m}{\partial \xi}$ $D-F\gamma_F(g)-B\gamma_B(g) = \xi \frac{df}{d\xi}$ (设这3个已知)

重整化群可以研究作用的尺度效应,

即在不同尺度下作用的强弱 $\rightarrow$

$\hookrightarrow$ 角解: $f(\xi) = \xi^D e^{-\int_1^\xi d\xi (F\gamma_F(\xi) + B\gamma_B(\xi)) / \xi}$

$g(\xi)$: 跑动耦合常数, $m(\xi)$: 跑动质量

QCD的 β 函数:

$$g = \mu^{-2/2} Z_1^{-1} Z_2 Z_3^{1/2} g_B = \mu^{-2/2} \left(1 - \frac{13}{3} \frac{\alpha_s}{2\pi\epsilon}\right)^{-1} \left(1 - \frac{4}{3} \frac{\alpha_s}{2\pi\epsilon}\right) \left(1 + \left(5 - \frac{2n_f}{3}\right) \frac{\alpha_s}{2\pi\epsilon}\right)^{1/2} g_B \approx \mu^{-2/2} \cdot \left(1 + \left(11 - \frac{2}{3}n_f\right) \frac{g^2}{16\pi^2\epsilon}\right)$$

$$\Rightarrow \beta(g) = \mu \frac{\partial g}{\partial \mu} = \lim_{\epsilon \rightarrow 0} \mu \frac{\partial g}{\partial \mu} = -\left(11 - \frac{2}{3}n_f\right) \frac{g^3}{16\pi^2}$$

若 $n_f \leq 16$, 则 $\beta < 0$, QCD渐近自由

渐近自由:



0与 g_0 为固定点.
 $\xi \rightarrow \infty$ 时, $g \rightarrow g_0$
($\beta > 0$, g 随 ξ 增)



$\beta < 0$, g 随 ξ 减
故 $\xi \rightarrow \infty$, $g \rightarrow 0$ (渐近自由, 即作用局域性)

耦合常数形式: $\alpha_s = \frac{g^2}{4\pi}$

$$\text{令 } \beta_0 = 11 - \frac{2}{3}n_f, \quad \mu \frac{\partial \alpha_s}{\partial \mu} = -\frac{\beta_0}{2\pi} \alpha_s^2 \Rightarrow \alpha_s = \frac{g^2(\mu)}{4\pi} = \frac{\alpha_s(\Lambda)}{1 + \frac{\beta_0}{4\pi} \alpha_s(\Lambda) \ln(\mu^2/\Lambda^2)}$$

$\mu \rightarrow \infty$ 时, $\alpha_s \rightarrow 0$; $\mu \rightarrow \Lambda$, α_s 强耦合

$$\approx \frac{4\pi}{\beta_0 \ln(\mu^2/\Lambda^2)} = \frac{4\pi}{(11 - \frac{2}{3}n_f) \ln(\mu^2/\Lambda^2)}$$

高阶修正: $\beta(g)$ 写作 g^2 级数

$$\frac{\mu}{\alpha_s} \frac{\partial \alpha_s}{\partial \mu} = \frac{2}{g} \beta(g) = -\beta_0 \left(\frac{\alpha_s}{2\pi}\right) - \beta_1 \left(\frac{\alpha_s}{2\pi}\right)^2 - \dots$$

$$\alpha_s(\mu) = \frac{4\pi}{\beta_0 \ln(\mu^2/\Lambda^2)} \left(1 - \frac{2\beta_1}{\beta_0^2} \frac{\ln(\ln(\mu^2/\Lambda^2))}{\ln(\mu^2/\Lambda^2)} + \frac{4\beta_2}{\beta_0^3 \ln(\mu^2/\Lambda^2)} \left((\ln(\ln(\mu^2/\Lambda^2)) - \frac{1}{2})^2 + \frac{\beta_2 \beta_0}{\beta_1^2} - \frac{5}{4} \right) + \dots \right)$$

$$\beta_1 = 51 - \frac{19}{3}n_f, \quad \beta_2 = \frac{1}{8}(2857 - \frac{5033}{9}n_f - \frac{325}{27}n_f^2) \dots$$

电子散射: n_f 取4

正负电子: n_f 取5

($n_f < 2$ 激发能 Q 的味数)